

Biomarker discovery in psoriasis: an integrated feature selection and machine learning framework for identifying diagnostic genes

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Background: Psoriasis is a chronic, immune-mediated disease characterized by inflammation and abnormal skin cell growth. The precise molecular mechanisms underlying its progression remain partially understood, highlighting the need to identify reliable gene biomarkers for early diagnosis and targeted therapies.

Methods: This study presents an integrated feature selection and machine learning framework to identify key genes associated with psoriasis. Gene expression profiles were collected and preprocessed from seven GEO databases. We evaluated various combinations of evolutionary algorithms for gene selection and machine learning classifiers. The Harris Hawks Optimization (HHO) algorithm combined with the XG Boost classifier was selected due to its superior performance in maximizing prediction accuracy.

Results: The hybrid HHO-XG Boost combination demonstrated superior performance, achieving an exceptional F1-score of 99.31% on the validation data. This optimized model successfully identified a set of 35 key genes highly predictive of psoriasis, including TNFRSF10C, MCOLN2, KYNU, and CD47.

Conclusion: The identified 35 genes have significant potential to serve as robust diagnostic and prognostic biomarkers for psoriasis. This study validates the effectiveness of the HHO-XG Boost integrated approach for biomarker discovery in complex diseases and provides strong candidates for future functional validation and targeted drug development.

Keywords: biomarkers, computational biology, gene expression, machine learning, psoriasis

Received: 1 September 2025
Accepted: 28 October 2025

Iran J Dermatol 2025; 28(Special Issue): 76-88

DOI: [10.22034/ijid.2025.544542.2091](https://doi.org/10.22034/ijid.2025.544542.2091)

What is already known on the subject?

- Investigating psoriasis through gene expression profiles is challenging due to the lack of a single, comprehensive dataset, and the limited

availability of small datasets. Previous studies have primarily relied on small datasets, employed machine learning methods exclusively, and utilized various feature extraction methods such

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Please cite this article as: Ansari Shiri R, Zare Mehrjardi F, Sardari Zarchi M, Ghaneai H. Biomarker discovery in psoriasis: an integrated feature selection and machine learning framework for identifying diagnostic genes. Iran J Dermatol. 2025; 28(Special Issue): 76-88.

as correlation analysis.

The study's main messages:

- Obtained an appropriate dataset by integrating multiple smaller datasets.
- Combining various machine learning methods with three evolutionary algorithms, we extract 35 genes associated psoriasis from approximately 54,000 genes.

INTRODUCTION

Psoriasis is a chronic, immune-mediated inflammatory skin disease affecting millions worldwide. Its complex etiology involves a combination of genetic and environmental factors, leading to the hyperproliferation of skin cells and the formation of characteristic erythematous plaques covered with silvery scales. Beyond its physical symptoms, psoriasis imposes a significant psychological and social burden on patients, often leading to stigma and a reduced quality of life. The disease is also associated with various comorbidities, including cardiovascular disease and metabolic syndrome, underscoring its systemic nature ¹.

Despite extensive research, a definitive cure for psoriasis remains elusive, and current treatments primarily focus on symptom management. Therefore, there is a critical need to deepen our understanding of the underlying molecular mechanisms of the disease to identify novel diagnostic biomarkers and therapeutic targets. Previous studies have employed various computational methods to analyze gene expression data, however challenges remain in effectively handling high-dimensional data and identifying the most relevant genes from a vast pool of candidates ¹.

This study aims to address these challenges by introducing a novel approach that combines single-objective evolutionary algorithms with machine learning methods. The primary objective is to leverage the power of evolutionary algorithms for feature selection to identify key genes from a large dataset, thereby enhancing the performance and interpretability of classification models. The most important genes are then used to develop a robust model for psoriasis diagnosis, which could serve as a valuable tool for early detection and personalized medicine ².

The identification of key genes and biomarkers for complex diseases such as psoriasis is a central

focus of modern biomedical research. Traditional methods often struggle with the high-dimensional nature of genomic data, promoting researchers to explore advanced computational approaches. In this context, bioinformatics and machine learning have emerged as powerful tools, with studies generally falling into two primary categories: those focusing on classification models and those developing novel feature selection methods, often incorporating metaheuristic algorithms ^{1,2}.

Several studies have successfully employed machine learning for disease diagnosis and classification. Ansari *et al.* ³ utilized gene expression profiles from four GEO databases to classify psoriasis patients. Their work demonstrated that by combining data preprocessing, normalization, and class balancing, the Support Vector Machine (SVM) algorithm achieved a high accuracy of 99.11%. The study also identified key genes such as CSF2, EPOR, D17GGCLE, and D15NDS2 as potential biomarkers. Similarly, Gu *et al.* ⁴ designed a classification model using an incremental feature selection algorithm. They identified only two genes, IGFL1 and C10orf99, which were sufficient to achieve a high accuracy of 99.81%. This finding highlights the potential of a small, highly effective gene set for creating robust predictive models.

Other researchers have focused on the critical step of feature selection. Saeed Azadifar *et al.* ⁵ proposed a multi-stage method for cancer diagnosis, which is also applicable to gene selection in other diseases. Their approach first filters genes using Fisher's exact test, then employs graph-based clustering with the Leven algorithm, and finally utilized a multi-objective particle swarm optimization algorithm (MaPSOGS) to select the final gene set. This method effectively reduced the number of genes while maintaining high classification accuracy. Kavitha *et al.* ⁶ also emphasized feature selection, using mutual information to identify important genes for a hybrid model (MKUSVM-RBFELM). Their approach achieved a high accuracy of 92.58% but did not include an in-depth biological analysis of the identified genes, highlighting a common limitation of some purely computational studies.

Beyond classification, machine learning has been employed to predict treatment responses and identify shared genetic signatures. Tsoi *et al.* ⁷ demonstrated that analyzing gene expression in non-lesional skin could predict a patient's response to anti-TNF therapy

with an Area Under Roc Curve (AUC) of 0.75, highlighting the USP18 gene as a key regulator. This finding underscores the potential of personalized medicine. In another study, Tian *et al.*⁸ developed an algorithm called GEE-TGDR to analyze longitudinal gene expression data, successfully identifying ten key lncRNAs associated with treatment response in psoriasis patients, achieving a prediction accuracy of 70%. Furthermore, Jing Zhang *et al.*⁹ utilized an integrated bioinformatics approach to identify eight shared key genes, including S100A9 and KYNU, between psoriasis and ulcerative colitis, demonstrating that common molecular pathways can exist across different inflammatory diseases.

While previous studies have made significant progress in using computational methods to understand psoriasis, many still face limitations, such as a reliance on specific datasets, limited generalizability, or a lack of deep biological interpretability. To address these gaps, the primary aim of this study is to propose and evaluate an integrated feature selection framework that combines the robust global search mechanisms of evolutionary algorithms with the high-performance classification strengths of machine learning methods. This hybrid approach, exemplified by the Harris Hawks Optimization (HHO) algorithm and the XG Boost classifier, seeks to improve upon previous methods by providing a more robust, accurate, and interpretable gene signature for psoriasis diagnosis and targeted therapy, thereby accurately identifying an optimal, minimal subset of diagnostic genes with superior performance.

METHODS

In this section, we first provide a brief introduction to the dataset and outline the steps of the method used

in this research, followed by a detailed description of the proposed approach.

Data Collection

The gene expression data utilized in this study were collected from seven reputable databases within the Gene Expression Omnibus (GEO) subset of the NCBI databases. These databases specifically contain genetic data related to various types of psoriasis, obtained from skin tissue samples of both patients and healthy individuals. The samples were categorized into two groups: the “patient” group, which include samples from lesional skin and the “healthy” group, which include samples from healthy individuals as well as non-lesional skin from patients.

To enhance the reliability and generalizability of the research, samples were collected from a diverse group of individuals with varying clinical conditions. In total, 1327 samples were obtained from these seven databases. The detailed distribution of healthy and patient samples across each database is presented in Table 1.

Data Pre-processing Steps

After collecting and merging the data from seven different databases, a comprehensive dataset was created, containing all samples and 53,952 common features. The following pre-processing steps were then applied to this dataset to ensure its quality and suitability for analysis.

Handling Invalid Values and Data Transformation

First, columns containing a high number of invalid values were removed. For columns with fewer missing values, an appropriate imputation strategy was applied to fill them. Additionally, all

Table 1. Details of the selected GEO datasets

Database Name	Number of samples from healthy individuals	Number of samples from sick individuals	Platform (GPL ID)
GSE106992	67	125	GPL570
GSE13355	122	58	GPL570
GSE78097	33	33	GPL570
GSE117468	242	323	GPL570
GSE55201	30	51	GPL570
GSE41664	53	53	GPL570
GSE0999	85	85	GPL96
All samples	444	673	

Abbreviations: GEO, Gene Expression Omnibus; GPL ID, GEO Platform Identifier; GSE, GEO Series (accession number for a study)

non-numeric features were identified and converted into numerical representations to ensure compatibility with machine learning models.

Data Normalization and Scaling

To mitigate large fluctuations in data values and align them to a consistent scale, a two-step normalization process was applied. First, a base-2 logarithmic transformation was performed. This common practice in bioinformatics helps stabilize variance and normalize the distribution of gene expression values. Second, robust statistical scaling was employed. Unlike standard scaling methods, this technique is less sensitive to outliers and improves the accuracy of probabilities assigned to extreme data points¹⁰.

Data Balancing

To address the imbalance between the number of healthy and patient samples, a hybrid technique combining SMOTE (Synthetic Minority Over-Sampling Technique) and Tomek Links was used. SMOTE generates synthetic samples for the minority class (healthy samples), while Tomek Links removes noisy and borderline samples near the decision boundary^{11,12}. This combined approach not only balances the dataset but also improves class separation, thereby enhancing the precision, recall, and F1 score of the machine learning models.

Data Partitioning

The final pre-processed dataset, comprising 1,327 samples and 53,952 features, was partitioned for

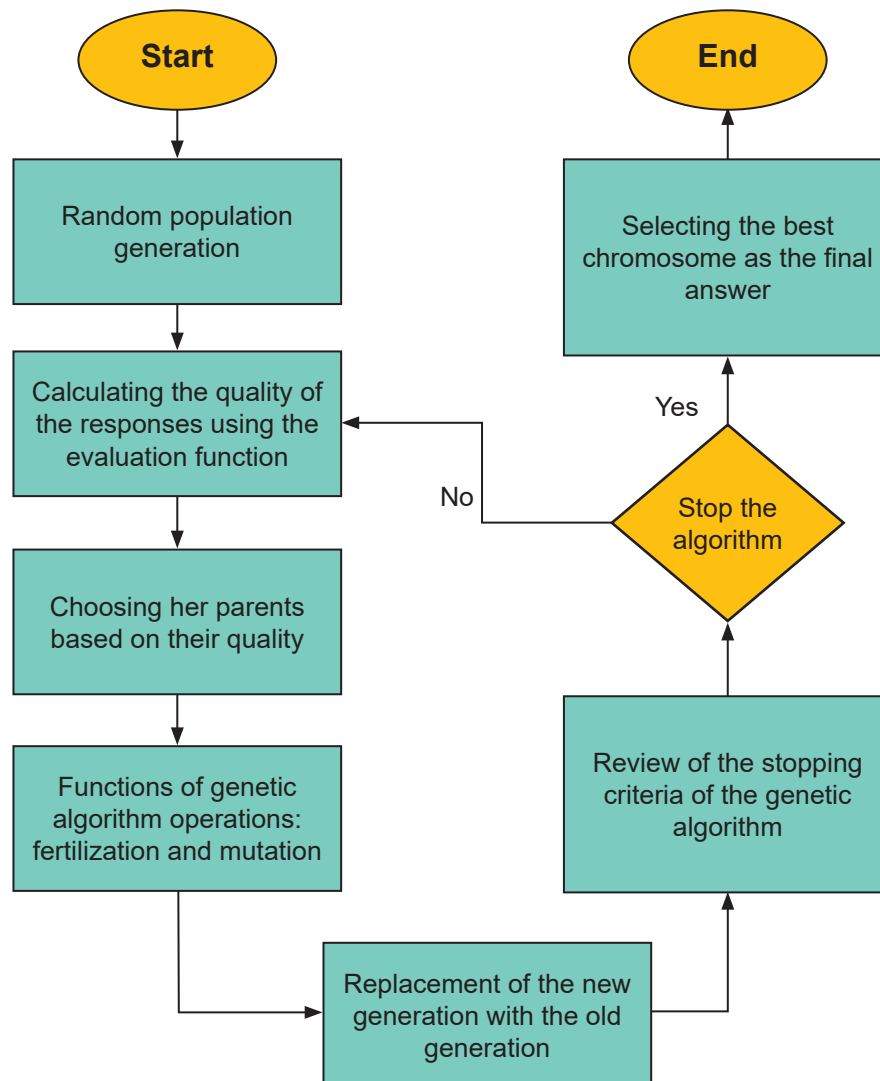


Figure 1. Flowchart of GA process in feature selection. GA, Genetic Algorithm

model training and evaluation. To ensure robust model validation, a 10-fold cross-validation method employed during the training and testing process.

Feature Selection using Evolutionary Algorithms

In this study, several evolutionary and meta-heuristic algorithms were evaluated for feature selection to identify the most relevant genes from a high-dimensional dataset. The following section provides a concise introduction to the algorithms considered.

Genetic Algorithm (GA)

The GA is a meta-heuristic inspired by the principle of natural selection. It operates on a population of candidate solutions, each represented as a chromosome (a subset of gene). Through operators such as selection, crossover, and mutation, the population evolves iteratively toward an optimal solution. This search is a guided random process that relies on a fitness function to evaluate the performance of each solution and direct the search toward promising regions of the problem space. GA is a robust method for solving complex combinatorial optimization problems and is particularly well-suited for gene selection tasks due to its ability to effectively explore a vast solution space^{13,14}. Figure 1 illustrates the steps of the GA in a flowchart.

Particle Swarm Optimization (PSO)

The PSO algorithm is a swarm intelligence method inspired by the social behavior of bird flocking or fish schooling. In PSO, each candidate solution is represented as a “particle” that moves through the search space. A particle’s movement is influenced by its own best-known position (p Best) and the best-known position of the entire swarm (g Best). PSO is known for its simplicity, fast convergence speed, and ease of implementation, as it does not require the calculation of derivative to be calculated. However, its primary limitations include a tendency to become trapped in local optima and a potential for poor performance in very high-dimensional problems^{15,16}. Figure 2 shows the steps of the PSO algorithm in a flowchart.

Harris Hawks Optimization (HHO)

The HHO algorithm is a meta-heuristic inspired

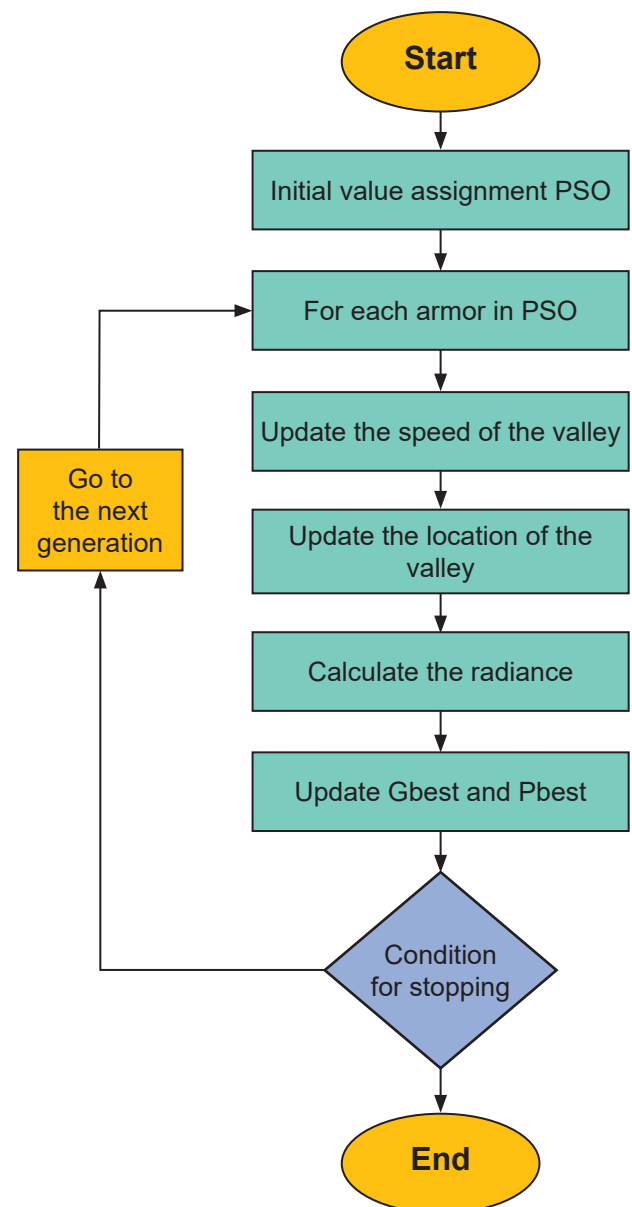


Figure 2. Flowchart of PSO Algorithm process in feature selection. PSO, Particle Swarm Optimization

by the collaborative hunting behavior of Harris’s hawks. The algorithm simulates the pursuit of prey through three main phases: exploration, transition, and exploitation. During the exploration phase, the hawks search for prey. The transition phase determines the shift from exploration to exploitation based on the prey’s escape energy. The exploitation phase employs different attack strategies -either soft or hard sieges- to capture the prey. HHO is notable for its excellent balance between exploration and exploitation, which helps prevent it from becoming trapped in local optima. This algorithm is highly

effective for a variety of complex optimization problems, including feature selection, and has demonstrated superior performance in many large-scale applications¹⁷. Figure 3 illustrates the steps of the HHO algorithm in a flowchart.

In this study, based on a comparative analysis of various methods, the HHO algorithm was selected as the primary feature selection method due to its demonstrated superior ability to effectively navigate the high-dimensional gene expression data and identify the optimal subset of genes with high classification performance.

Data Classification using Machine Learning Models

Following the feature selection process, the classification of healthy and patient samples was conducted using machine learning models. The primary classifier in this study was the XG Boost (Extreme Gradient Boosting) algorithm, selected for its superior performance, speed, and ability to handle high-dimensional data. To provide a comparative

analysis and to demonstrate the robustness of our approach, several other widely-used machine learning models were also evaluated. A brief overview of the classification models employed is provided below:

Support Vector Machine (SVM): This method seeks to identify the optimal separating hyperplane that maximizes the margin between different classes, thereby enhancing the model's generalization ability.

Random Forest: As an ensemble learning method, Random Forest constructs numerous of decision trees during training. The final classification is determined by a majority vote among the individual trees, which helps reduce overfitting and improve accuracy.

Decision Tree: This is a non-parametric supervised learning method that construct a model in the form of a tree structure. It makes a series of decisions based on feature values to iteratively partition the dataset until data points are correctly assigned to their respective leaf nodes.

AdaBoost: This adaptive boosting algorithm combines multiple weak classifiers into a single strong one. It works by iteratively training new models to

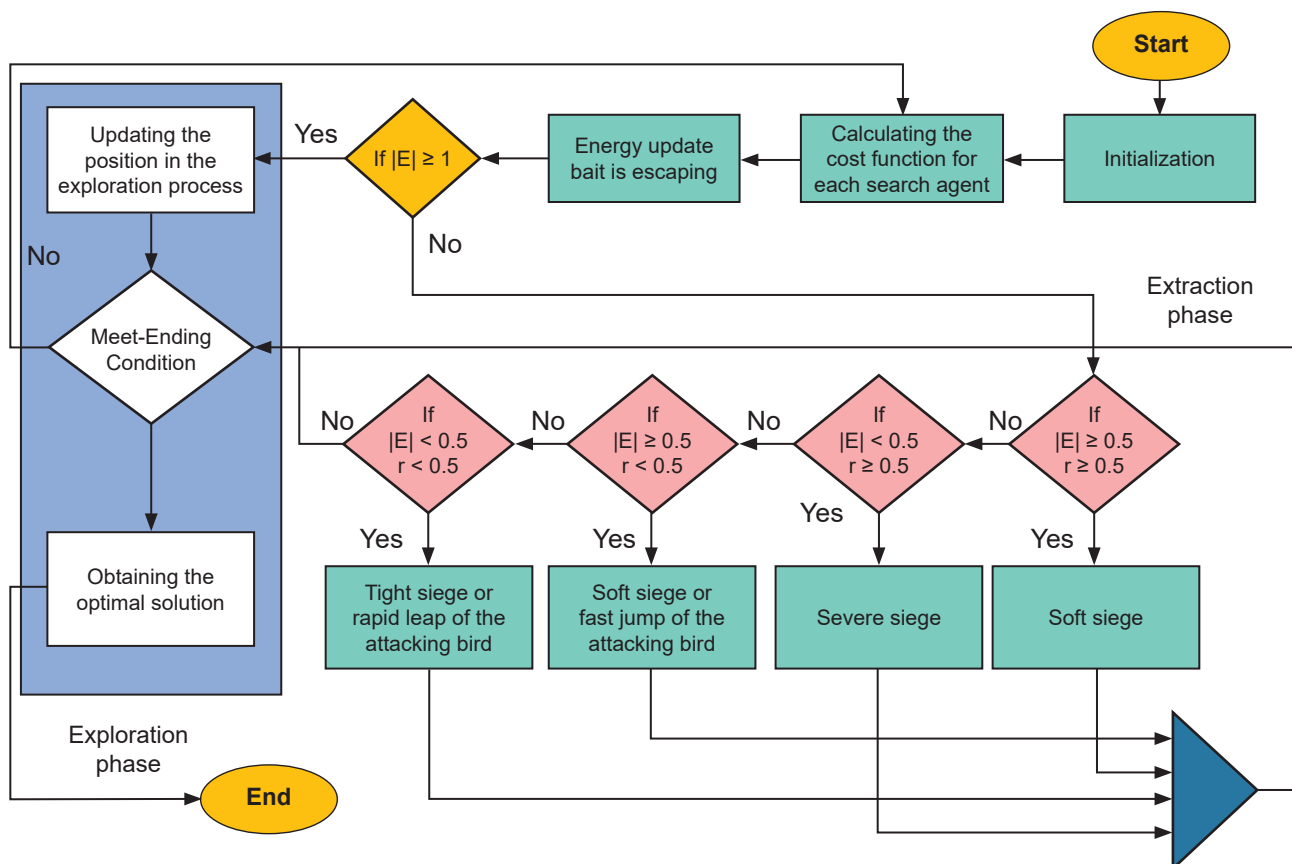


Figure 3. Flowchart of HHO Algorithm process in feature selection. HHO, Harris Hawks Optimization

correct the misclassifications of previous models, placing greater emphasis on data points that were previously difficult to classify.

Gradient Boosting: A powerful ensemble technique, Gradient Boosting builds a strong predictive model by sequentially adding weak learners. Each new model is trained to minimize the errors of the preceding models, which leads to a highly accurate final classifier ¹².

Ethical Consideration

This study is a secondary analysis of publicly available gene expression data obtained from the Gene Expression Omnibus (GEO) repository. All datasets used were de-identified and complied with the ethical standards and data-sharing policies of the original studies, which had obtained the necessary ethical approvals and patient consents. Therefore, no additional ethical approval was required for this computational analysis.

RESULTS

In this study, the performance of the proposed model for psoriasis diagnosis based on gene expression data was evaluated. The methodology comprised several key steps: data collection from seven GEO databases, pre-processing including the removal of invalid values, data normalization and balancing using a combination of SMOTE and Tomek Links, and finally, partitioning the data into training (90%) and test (10%) sets using 10-fold cross-validation. Subsequently, the results of feature selection using evolutionary algorithms and classification with machine learning models were analyzed. The parameters of the three evolutionary algorithms are presented in Table 2. It should be noted that all evolutionary algorithms were developed under identical conditions, with the same population size, number of iterations, and random initial population.

To evaluate the effectiveness of the proposed approach, the performance of various machine learning models combined with three single-objective feature selection algorithms -GA, HHO, and PSO- was compared. The primary evaluation metric was the F1-score, which calculates the harmonic mean of precision and recall, providing a balanced measure for assessing models on imbalanced data ¹⁸. To evaluate the optimization efficiency, Figure 4

Table 2. Parameters of the three evolutionary algorithms

Algorithm name	Parameters	Values
GA	Population Size	30
	Number of iteration	200
	Crossover Rate	0.85
	Mutation Rate	0.1
PSO	Number of Particles	30
	Number of iteration	200
	Cognitive Coefficient (c1)	2
	Social Coefficient (c2)	2
HHO	Inertia Coefficient (w)	0.4-0.9
	Number of Hawks	30
	Number of iteration	200

Abbreviations: GA, Genetic Algorithm; HHO, Harris Hawks Optimization; PSO, Particle Swarm Optimization

specifically illustrates the convergence behavior of the three evolutionary algorithms (HHO, GA, and PSO) when exclusively paired with the superior XG Boost classifier. Conversely, Figure 5, presents a comprehensive performance comparison by depicting the F1-score evaluation of all possible hybrid combinations, unequivocally confirming the unrivaled dominance of the HHO-XG Boost hybrid method.

The results demonstrated that the HHO algorithm, combined with the XG Boost classifier, achieved the highest performance, attaining an F1-score of 99.31%. This finding indicates that the gene set selected by HHO is optimal for distinguishing between healthy and patient samples. The detailed average performance across five independent runs for all model combinations is presented in Table 3.

The results in Table 3 clearly demonstrate that among the three feature selection algorithms, HHO combined with XG Boost exhibited superior performance in psoriasis diagnosis, and achieving the highest accuracy.

As explained in the Methods section, the primary goal of this research was to identify an optimal subset of genes. Using the HHO algorithm in conjunction with the XG Boost classifier, 35 key genes were identified that showed the highest discriminatory power between healthy and patient samples. These genes can serve as biomarkers for early diagnosis and as therapeutic targets for designing personalized psoriasis treatments. Table 4 presents the list of Identified Key Genes.

In summary, the results of this study successfully demonstrate the efficacy of the hybrid HHO-XGBoost

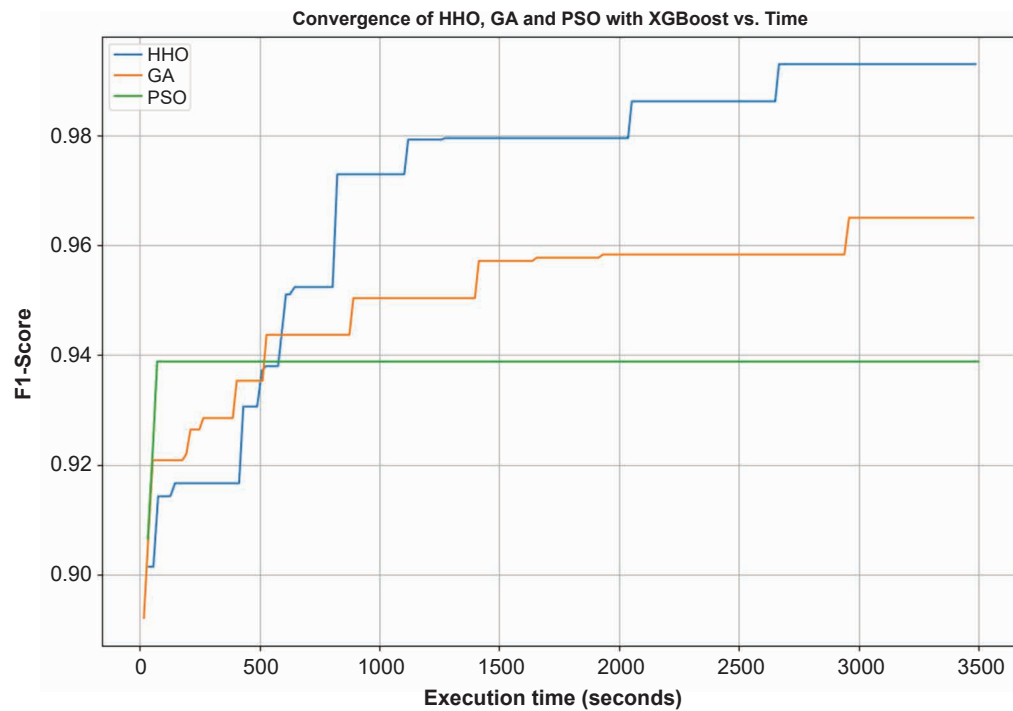


Figure 4. Convergence diagram of three evolutionary algorithms with the XGBoost method. GA, Genetic Algorithm; HHO, Harris Hawks Optimization; PSO, Particle Swarm Optimization; XGBoost, Extreme Gradient Boosting

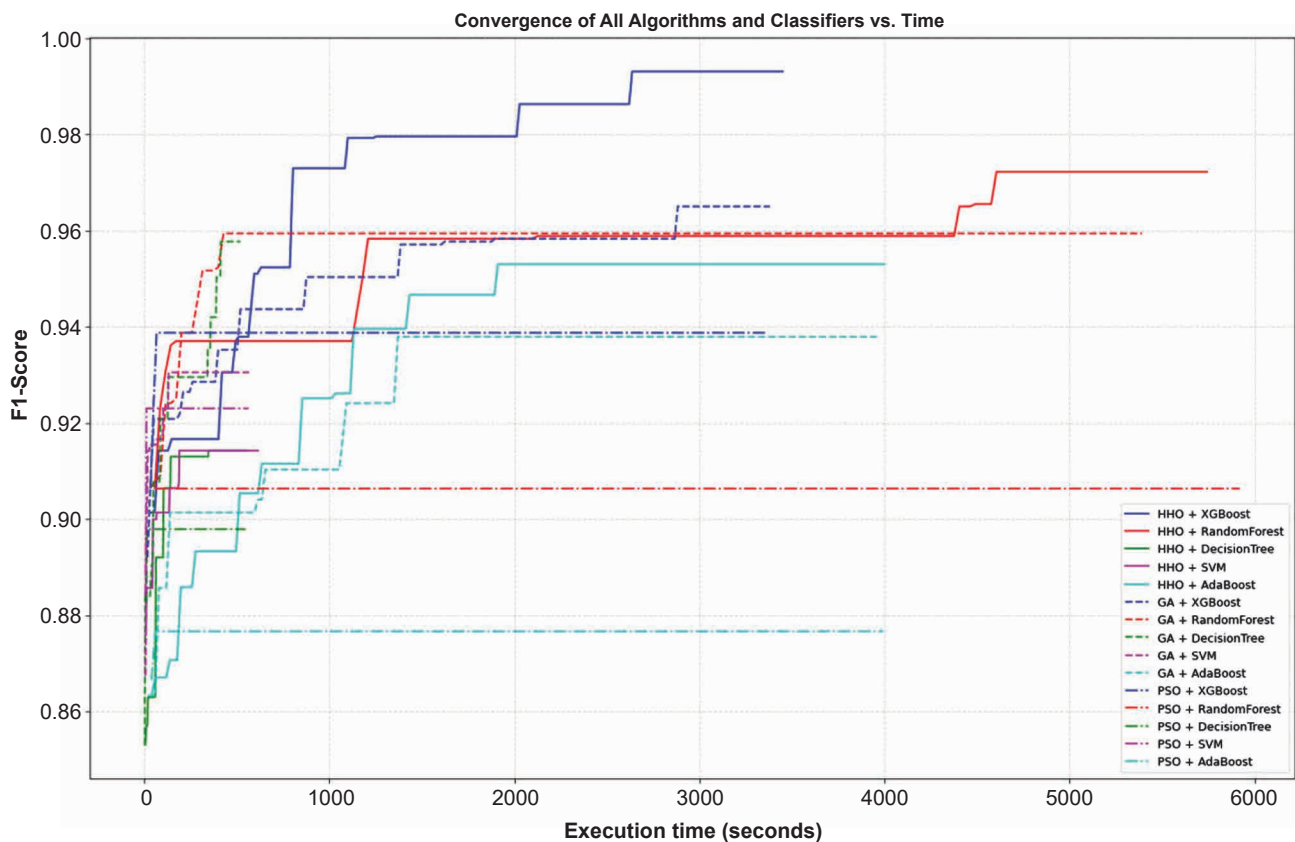


Figure 5. F1-score evaluation of all hybrid feature selection and classification combinations (single run). AdaBoost, Adaptive Boosting; GA, Genetic Algorithm; HHO, Harris Hawks Optimization; PSO, Particle Swarm Optimization; SVM, Support Vector Machine; XGBoost, Extreme Gradient Boosting

Table 3. Comparison of Classification Model Performance with Various Feature Selection Algorithms (based on F1-score)

Feature Selection Algorithm	Decision Tree	AdaBoost	SVM	Random Forest	XG Boost
GA	93.62%	92.31%	93.62%	95.95%	96.50%
HHO	91.67%	95.30%	93.62%	97.22%	99.31%
PSO	92.09%	91.78%	92.96%	92.31%	93.88%

Abbreviations: AdaBoost, Adaptive Boosting; GA, Genetic Algorithm; HHO, Harris Hawks Optimization; PSO, Particle Swarm Optimization; SVM, Support Vector Machine; XGBoost, Extreme Gradient Boosting

Table 4. List of Identified Key Genes form proposed method

Gene Symbol	Gene Name	General Biological Function	Relevance to Psoriasis	Key Evidence / Pathway Link
TNFRSF10C	TNF receptor superfamily member 10c	Decoy Receptor for TRAIL; modulates apoptosis.	Direct	Immune Dysregulation; essential for keratinocyte survival and homeostasis under inflammatory conditions ⁴ .
MCOLN2	mucolipin TRP cation channel 2	Cation channel; involved in endosomal/lysosomal trafficking.	Indirect	Vesicle Transport; affects the secretion of inflammatory cytokines and endosomal processing in immune cells ^{4,5} .
KYNU	Kynureninase	Enzyme in the Tryptophan catabolic pathway.	Direct	Metabolic Stress; Kynurenine metabolites are psychoactive/inflammatory and link gut-skin axis dysregulation to psoriasis inflammation ^{2,9} .
CD47	CD47 molecule	Integrin-associated protein; "Don't-eat-me" signal.	Direct	Immune Evasion; regulates macrophage phagocytosis and contributes to the persistence of damaged cells in psoriatic plaques ^{3,7} .
TBX5-AS1	TBX5 antisense RNA 1	Long non-coding RNA (lncRNA); transcriptional regulator.	Indirect	Transcriptional Modulation; lncRNAs play a role in regulating the expression of genes involved in skin inflammation and proliferation ⁸ .
TEAD4	TEA domain transcription factor 4	Nuclear transcription factor; component of the Hippo signaling pathway.	Direct	Cell Proliferation; a known driver of uncontrolled cell growth and hyperproliferation characteristic of psoriasis ^{4,9} .
FCRLA	Fc receptor like A	Fc receptor-like protein in B cells.	Indirect	Immune Response; secondary player in B cell development and the chronic adaptive immune response in psoriasis ^{3,7} .
MDGA2	MAM domain containing... anchor 2	GPI-anchored glycoprotein; cell adhesion/synaptic organization.	Indirect	Adhesion/Structural; contributes to cell-cell interactions and differentiation processes in the rapidly changing epidermis ⁴ .
ZAN	Zonadhesin	Protein primarily involved in fertilization.	Secondary/General	General Cellular Machinery; a highly conserved gene that may participate in an as-yet-undiscovered regulatory or structural mechanism in the skin ⁵ .
DNAAF4	dynein axonemal assembly factor 4	Component of the cilia/flagella assembly complex.	Secondary/General	General Cellular Machinery; its selection may indicate general cellular stress affecting core protein complexes ⁵ .
C2orf74-DT	C2orf74 divergent transcript	Intergenic non-coding RNA (lncRNA).	Secondary/General	Unknown Function; likely a non-specific regulatory element responding to the inflammatory state ⁴ .
PDLIM7	PDZ and LIM domain 7	Scaffold protein for Actin Filaments.	Indirect	Cytoskeletal Organization; relevant to keratinocyte morphology, migration, and maintenance of the epithelial barrier ⁴ .
PRRX1	paired related homeobox 1	Homeobox transcription factor; tissue differentiation.	Indirect	Tissue Differentiation; regulates gene expression critical for proper skin development and maintenance ⁴ .
FXR2	FMR1 autosomal homolog 2	RNA-binding protein (RBP); regulates mRNA translation.	Indirect	RNA Regulation; controls the fate of specific mRNAs, potentially regulating the expression of inflammatory mediators ⁵ .
MRTFB	Myocardin related transcription factor B	Transcriptional activator related to muscle/smooth muscle differentiation.	Indirect	Cell Differentiation; may influence the differentiation and function of stromal cells (fibroblasts) in the inflamed dermis ⁵ .
FZR1	fizzy and cell division cycle 20 related 1	APC/C complex activator; regulates the Cell Cycle.	Direct	Cell Cycle Control; directly linked to the uncontrolled and accelerated cell cycle progression (hyperproliferation) in psoriasis keratinocytes ⁶ .

Table 4. Continued

Gene Symbol	Gene Name	General Biological Function	Relevance to Psoriasis	Key Evidence / Pathway Link
MCCC2	Methylcrotonyl-CoA carboxylase subunit 2	Enzyme in Leucine oxidation.	Indirect	Metabolism; part of amino acid metabolism. High expression reflects the heightened metabolic demands of rapidly dividing cells ⁵ .
RPLP2	Ribosomal protein lateral stalk subunit P2	Component of the 60S ribosomal subunit.	Secondary/ General	Protein Synthesis; reflects the high protein synthesis demand and general cellular stress in hyperproliferative tissue ⁵ .
GSDMD	Gasdermin D	Pore-forming protein; primary mediator of inflammatory cell death (Pyroptosis).	Direct	Inflammation/Cell Death; crucial in mediating the inflammatory loop by releasing cytokines (IL-1 β , IL-18) ² .
LINC00334	Long intergenic non-protein coding RNA 334	Intergenic non-coding RNA (lncRNA).	Secondary/ General	Unknown Function; likely a non-specific regulatory element responding to the inflammatory state ⁴ .
PEX10	Peroxisomal biogenesis factor 10	Involved in protein import into Peroxisomes.	Indirect	Peroxisome Biogenesis; linked to lipid metabolism and oxidative stress, both relevant to the psoriasis microenvironment ⁴ .
CASTOR3P	CASTOR family member 3, pseudogene	Pseudogene; linked to Arginine sensing and TORC1 signaling.	Indirect	Metabolic Signaling; Nutrient sensing pathways (TORC1) are vital for regulating cell growth and proliferation in psoriasis ⁵ .
VSTM2L	V-set and transmembrane domain containing 2 like	Antagonist of neuroprotective Humanin.	Indirect	Apoptosis/Survival; may be involved in general cell survival or anti-apoptotic mechanisms that favor hyperproliferation ⁴ .
PPWD1	Peptidylprolyl isomerase domain and WD repeat containing 1	Catalytic subunit of the spliceosome.	Secondary/ General	Splicing/Protein Folding; reflects the general increased cellular activity and high processing demands in the inflamed tissue ⁵ .
SOWAHC	Sosondowah ankyrin repeat domain family member C	Associated with Rho GTPase signaling.	Indirect	Cytoskeletal/Signaling; Rho pathways regulate keratinocyte morphology, migration, and stress fibers ⁴ .
PARD6B	par-6 family cell polarity regulator beta	Regulates Cell Polarity (PDZ/CRIB domains).	Indirect	Cell Polarity; dysregulation contributes to the loss of tissue architecture and epithelial function in the psoriatic lesion ⁴ .
TNPO3	Transportin 3	Nuclear transporter for splicing factors.	Secondary/ General	Nuclear Transport; reflects the high transcriptional and splicing demands of rapidly dividing psoriasis cells ⁵ .
PLCD3	Phospholipase C Delta 3	Increases intracellular Ca ²⁺ (via DAG and IP ₃)	Indirect	Signal Transduction; Ca ²⁺ signaling is a primary regulator of keratinocyte differentiation, which is aberrant in Pso ⁴ .
HSD17B14	Hydroxysteroid 17-beta dehydrogenase 14	Enzyme in steroid and lipid metabolism.	Indirect	Steroid/Lipid Metabolism; modifies local lipid and hormone signals, crucial for skin barrier function and inflammation ⁴ .
SMARCD2	SWI/SNF related BAF chromatin remodeling complex subunit D2	Subunit of the SWI/SNF Chromatin Remodeling Complex.	Indirect	Gene Regulation; affects chromatin structure and regulates the transcription of inflammatory and differentiation-related genes ⁴ .
ADGRL3	Adhesion G protein-coupled receptor L3	G protein-coupled receptor; cell adhesion.	Indirect	Adhesion/GPCR Signaling; mediates external signals and influences cell-cell interactions in the dermal-epidermal junction ⁵ .
H6PD	Hexose-6-phosphate dehydrogenase	Enzyme in hexose-phosphate metabolism.	Indirect	Metabolism; contributes to overall energy/ metabolism in the skin, which is altered by the inflammatory state ⁵ .
PRKAA2	Protein kinase AMP-activated catalytic subunit alpha 2	Catalytic subunit of AMPK; cellular energy sensor.	Direct	Metabolic Stress; key anti-inflammatory and metabolic regulator; its dysfunction drives psoriasis hyperproliferation ^{4,5} .
GCLC	Glutamate-cysteine ligase catalytic subunit	Rate-limiting enzyme in Glutathione synthesis.	Direct	Oxidative Stress; controls the cellular antioxidant capacity, which is severely compromised in psoriatic lesions ^{4,5} .
FN1	Fibronectin 1	Extracellular Matrix (ECM) protein; adhesion, migration, wound healing.	Direct	Tissue Remodeling; prominent in psoriatic plaques; involved in cell migration and the wound-healing characteristics of psoriasis ^{4,8} .

approach in accurately and efficiently identifying key genes associated with psoriasis. This model not only achieved an exceptionally high predictive accuracy (F1-score: 99.31) but also isolated a crucial set of genes with significant potential for use in future biological research and the development of personalized medicine. The detailed functional analysis presented in Table 4 confirms the biological relevance of all 35 genes identified by the optimal algorithm. These genes were initially selected through a robust, data-driven computational framework designed to minimize feature redundancy and maximize predictive power. This automated selection successfully identified genes that collectively regulate three central pillars of psoriasis pathogenesis: immune dysregulation (e.g., CD47), altered cell survival and proliferation (e.g., GSDMD, FZR1), and metabolic stress response (e.g., PRKAA2, GCLC). This strong alignment with established biological pathways validates the superiority of our evolutionary feature selection approach, firmly establishing these 35 genes as high-confidence biomarkers for early diagnosis and targeted therapeutic development in psoriasis.

DISCUSSION

In light of the current investigation and the comprehensive analysis of the results, the 35-gene panel identified by the optimized HHO-XGBoost algorithm provides a robust and highly predictive molecular signature for psoriasis. This gene set was primarily derived through a powerful, data-driven computational framework designed to maximize predictive accuracy by efficiently navigating a

high-dimensional feature space. The selection was subsequently subjected to rigorous genetic and bioinformatics validation to confirm its mechanistic relevance, as detailed in Table 4. Our analysis revealed that the 35 selected genes are not merely statistical artifacts but cluster into distinct functional categories, offering a biologically comprehensive perspective on psoriasis pathogenesis, as summarized in Table 5.

The panel successfully identified nine established Core Genes -including CD47, GSDMD, GCLC, and PRKAA2- that have direct, confirmed roles in immune evasion, pyroptotic inflammation, metabolic stress, and uncontrolled proliferation. Importantly, the algorithm also incorporated 19 modulatory genes, such as MCOLN2, PARD6B, and PLCD3 which regulate secondary cellular functions like cytoskeletal dynamics and signaling; their altered expression is a direct consequence of the inflammatory state. The consistency of this multi-tiered selection with published literature strongly validates the superior feature selection capability of our hybrid evolutionary approach ^{2,9}. Furthermore, the inclusion of seven secondary/general genes (e.g., RPLP2 and TNPO3), which are associated with general cellular machinery, is significant, as their dysregulation underscores the substantial overall burden and cellular stress imposed by the rapid proliferation characteristic of the psoriatic microenvironment. By integrating these direct drivers with their critical secondary modulators, this 35-gene set provides a highly comprehensive and biologically validated signature, establishing them as high-confidence candidates for personalized diagnostic and therapeutic strategies in psoriasis.

Table 5. Categorical summary of Gene set validation and relevance (Revised with Exact Counts)

Relevance category	Exact Gene Count	Role in psoriasis pathogenesis (Pathway link)	Key examples
Core Genes	9 Genes	Genes directly involved in primary disease mechanisms (inflammation, immunity, inflammatory cell death, and uncontrolled proliferation), widely validated in published literature.	TNFRSF10C, KYNU, CD47, TEAD4, FZR1, GSDMD, PRKAA2, GCLC, FN1
Modulatory Genes	19 Genes	Genes indirectly influencing cellular function (e.g., cytoskeletal structure, protein trafficking, secondary metabolism) whose expression is altered by the inflammatory and stress state.	MCOLN2, TBX5-AS1, FCRLA, MDGA2, PDLIM7, PRRX1, FXR2, MRTFB, MCCC2, PEX10, CASTOR3P, VSTM2L, SOWAHC, PARD6B, PLCD3, HSD17B14, SMARCD2, ADGRL3, H6PD
Secondary/general Genes	7 Genes	Genes with fundamental and generalized cellular roles (e.g., protein translation, nuclear transport). Their presence signifies the high overall stress and cellular turmoil resulting from rapid proliferation.	ZAN, DNAAF4, C2orf74-DT, RPLP2, LINC00334, PPWD1, TNPO3

CONCLUSION

Psoriasis is a chronic autoimmune disease for which no definitive cure currently exists, making the precise identification of its biomarkers crucial for research¹. This study presents a novel approach that combines evolutionary algorithms and machine learning to identify genes associated with the disease. After collecting and pre-processing gene expression data from seven GEO databases, the XGBoost model, combined with the HHO feature selection algorithm, demonstrated the best performance. The model achieved an impressive F1-score of 99.31% and identified 35 key genes out of the 53,952 genes examined. The primary strength of this method lies in its ability to select a small yet highly accurate subset of genes from a large, complex dataset, ensuring that the selected features are the most statistically relevant. Bioinformatics analysis confirmed that these genes are intricately linked to the core pathways of psoriasis, including immune, metabolic, and hyperproliferative pathways. However, as is common in similar studies^{5,6}, the primary challenge remains the need for further experimental validation of the identified genes using independent wet-lab techniques, as well as testing on new, external datasets to confirm generalizability and robustness across different populations. These findings hold significant potential for use as biomarkers for early diagnosis and as therapeutic targets in the development of targeted drugs for this disease. For future work, it is recommended to employ multi-objective optimization algorithms be used for gene selection to simultaneously optimize criteria such as model complexity, biological relevance, and classification accuracy. Incorporating more diverse datasets could further enhance generalizability and robustness of the model. This research represents a significant step toward developing more accurate and reliable diagnostic and therapeutic tools.

Acknowledgment

This research utilized AI tools for grammar and spelling corrections. All scientific and research content was prepared by the authors, with the AI tool serving solely as an assistant to enhance the writing. The authors acknowledge the AI's assistance in reviewing and refining the final text.

Authors' contributions

R.A. Shiri collected the data, implemented the

proposed method, and drafted manuscript. F.Z. Mehrjardi contributed to method implementation, analyzed the results, and participated in writing the manuscript. M.S. Zarchi contributed to manuscript review and editing. H. Ghaneai contributed to manuscript review and editing.

Funding source

The authors received no financial support for the research, authorship, or publication of this article.

Conflict of interest: None declared.

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